

[Back To Practice](#)[Save](#)

LRU Page Replacement Algorithm

Completed

Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits.

Input Format

- The first line of input contains an integer representing the number of pages in the reference string.
- The second line contains the page reference string as a sequence of integers.
- The third line contains an integer representing the number of available frames.

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, f, i, j, k, pos, faults = 0, hits = 0, count = 0;
5
6     printf("Enter number of pages in reference string: \n");
7     scanf("%d", &n);
8
9     int ref[50];
10    printf("Enter the reference string: \n");
11    for(i = 0; i < n; i++)
12        scanf("%d", &ref[i]);
13
14    printf("Enter number of frames: \n");
15    scanf("%d", &f);
16
17    int frame[10], time[10];
18
19    for(i = 0; i < f; i++) {
20        frame[i] = -1;
21        time[i] = 0;
22    }
23
24    for(i = 0; i < n; i++) {
```

Test Cases 2

[Run Sample Cases](#)[Run All Cases](#)

FACE Prep

Operating System - Coding | REC | 3 Feb

Course Progress80%Go Back

Back To Practice

Q1Save

Sample Input 1

5

1 2 3 4 5

5

Sample Output 1

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Total Page Faults: 5

Total Page Hits: 0

How it works

Step	Page	Frame State
1	1	1 - - - -

Select Language : C (GCC 9.2.0)

```
23
24     for(i = 0; i < n; i++) {
25         int found = 0;
26
27         for(j = 0; j < f; j++) {
28             if(frame[j] == ref[i]) {
29                 hits++;
30                 count++;
31                 time[j] = count;
32                 found = 1;
33                 break;
34             }
35         }
36
37         if(!found) {
38             pos = 0;
39             for(j = 1; j < f; j++)
40                 if(time[j] < time[pos])
41                     pos = j;
42
43             frame[pos] = ref[i];
44             count++;
45             time[pos] = count;
46             faults++;
47         }
48     }
```

Test Cases 2

Run Sample CasesRun All Cases

FACE Prep

Operating System - Coding | REC | 3 Feb

Course Progress80%Go Back

Back To Practice

Total Page Faults: 5

Total Page Hits: 0

How it works

Step	Page	Frame State
1	1	1 - - - -
2	2	1 2 - - -
3	3	1 2 3 - -
4	4	1 2 3 4 -
5	5	1 2 3 4 5

Q1

```
37
38     if(!found) {
39         pos = 0;
40         for(j = 1; j < f; j++)
41             if(time[j] < time[pos])
42                 pos = j;
43
44         frame[pos] = ref[i];
45         count++;
46         time[pos] = count;
47         faults++;
48     }
49
50     printf("\nTotal Page Faults: %d", faults);
51     printf("\nTotal Page Hits: %d", hits);
52
53     return 0;
54 }
```

Test Cases 2

Run Sample CasesRun All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

33°C Mostly clear

Search

ENG IN

19:08 17-05-2026

FACE Prep

Operating System - Coding | REC | 3 Feb

Course Progress78%Go Back

Back To Practice

Q1

Save

Total Page Faults: 5

Total Page Hits: 0

How it works

Step	Page	Frame State
1	1	1 - - - -
2	2	1 2 - - -
3	3	1 2 3 - -
4	4	1 2 3 4 -
5	5	1 2 3 4 5

Custom Test Case

Custom Test

Success

Input:

5
1 2 3 4 5
5

Expected Output:

Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0

Output:

Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0

Sample Test Case

> Test Case - 1

Success

33°C
Mostly clear

Search

ENG
IN

19:09
17-05-2026